

Applications ML Assignment2_Telco...

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```
[72]:
```

```
    ],  
    "MCC Score": [  
        lr_results[5],  
        dt_results[5],  
        knn_results[5],  
        nb_results[5],  
        rf_results[5]  
    ]  
})  
  
results
```

	Model	Accuracy	AUC Score	Precision	Recall	F1 Score	MCC Score
0	Logistic Regression	0.799148	0.840280	0.642633	0.548128	0.591631	0.462120
1	Decision Tree	0.730305	0.662526	0.492386	0.518717	0.505208	0.320220
2	KNN	0.742370	0.760513	0.515068	0.502674	0.508796	0.334273
3	Naive Bayes	0.746629	0.820104	0.516008	0.732620	0.605525	0.441302
4	Random Forest	0.792051	0.822504	0.637288	0.502674	0.562033	0.433354

Observations Five classification algorithms were evaluated for predicting customer churn:

1. Logistic Regression
2. Decision Tree Classifier
3. K-Nearest Neighbor Classifier
4. Gaussian Naive Bayes Classifier
5. Random Forest Ensemble Model

Logistic Regression achieved the highest accuracy (79.91%), highest AUC score (0.8403), and highest performance.

Random Forest achieved similar accuracy (79.21%) but slightly lower AUC and MCC values.

Naive Bayes obtained the highest recall (73.26%) and F1-score (60.55%), making it effective for identifying churned customers.

Decision Tree and KNN produced comparatively lower performance across most metrics.

Therefore, Logistic Regression was selected as the final model for deployment due to its balanced performance.

Best Model: Logistic Regression

Accuracy : 79.91% AUC Score: 0.8403 Precision: 64.26% Recall : 54.81% F1 Score : 59.16% MCC Score: 0.4621

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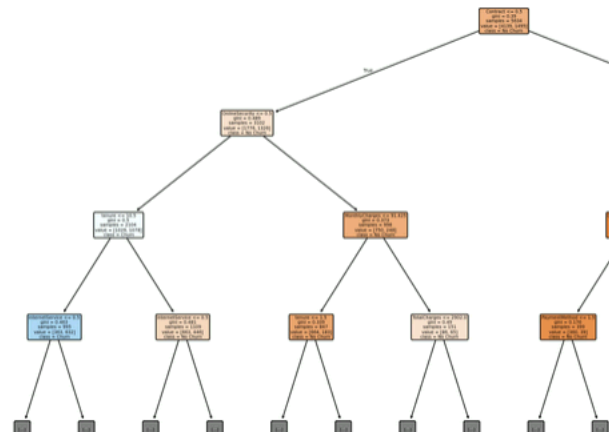
Markdown

```
plt.figure(figsize=(20,10))

tree.plot_tree(
    dt,
    feature_names=X.columns,
    class_names=["No Churn", "Churn"],
    filled=True,
    rounded=True,
    max_depth=3
)

plt.title("Decision Tree Visualization")
plt.show()
```

Decision Tree Visualization



```
[104]: from sklearn.metrics import roc_curve, auc

# -----
# ROC Curve Comparison
# -----

plt.figure(figsize=(8,6))
```

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Step 13: Compare Results

```
[72]: results = pd.DataFrame({
    "Model": [
        "Logistic Regression",
        "Decision Tree",
        "KNN",
        "Naive Bayes",
        "Random Forest"
    ],
    "Accuracy": [
        lr_results[0],
        dt_results[0],
        knn_results[0],
        nb_results[0],
        rf_results[0]
    ],
    "AUC Score": [
        lr_results[1],
        dt_results[1],
        knn_results[1],
        nb_results[1],
        rf_results[1]
    ],
    "Precision": [
        lr_results[2],
        dt_results[2],
        knn_results[2],
        nb_results[2],
        rf_results[2]
    ],
    "Recall": [
        lr_results[3],
        dt_results[3],
        knn_results[3],
        nb_results[3],
        rf_results[3]
    ],
    "F1 Score": [
        lr_results[4],
        dt_results[4],
        knn_results[4],
        nb_results[4],
        rf_results[4]
    ],
    "MCC Score": [
        lr_results[5],
        dt_results[5],
        knn_results[5],
        nb_results[5],
        rf_results[5]
    ]
})
```

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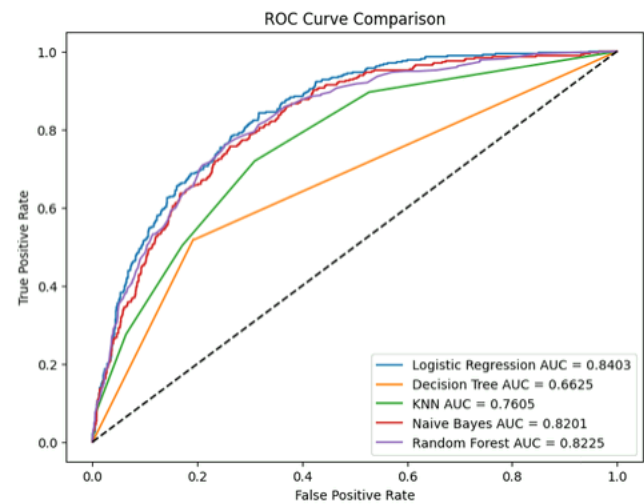
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 Code

```
plt.plot(fpr, tpr, label=model_name + " AUC = " + str(round(roc_auc, 4)))  
plt.plot([0, 1], [0, 1], "k--")  
plt.xlabel("False Positive Rate")  
plt.ylabel("True Positive Rate")  
plt.title("ROC Curve Comparison")  
plt.legend()  
plt.show()
```



```
[66]: # -----  
# Save Test Data for Streamlit  
# -----  
  
test_data = X_test.copy()  
test_data["Churn"] = y_test.values  
  
test_data.to_csv("test_data.csv", index=False)  
  
print("test_data.csv saved successfully")  
  
test_data.csv saved successfully
```